



Fig. 17. Search latency on different throughputs. The search latency of RAG systems [53] is also shown on the figure (red dashed line).

saturate the bandwidth of less than 1 SSD, so we will use 1 SSD for a fair comparison. We also show the cost-effectiveness of the ideal system mentioned in §3.1 (“Ideal” in the table). According to the estimation, the ideal system can saturate 3.6 SSDs, so 4 SSDs will be used.

- (3) **CPU+GPU (BANG, RUMMY).** To support a billion-scale dataset, larger DRAM is needed. A maximum of 1 TB DRAM is needed for DEEP1B. No SSDs are required.

The comparison of the three setups is shown in Table 1. GUSTANN (GPU+CPU+SSD) is 5.09× more cost-effective than CPU+SSD system (1.74× than the ideal CPU+SSD system), and 2.62× more cost-effective than CPU+GPU system. By introducing GPU, GUSTANN achieves a much higher search throughput compared to the CPU+SSD system. GUSTANN also prevents the non-scalable DRAM in CPU+GPU systems by using high-throughput SSDs.

5.6 Latency Analysis

GUSTANN trades latency off for throughput, because of the large-scale batched query. We measure the search latency against search throughput by adjusting the number of minibatches and number of queries processed per minibatch. We use SIFT1B dataset and show the results for recall@10=0.9. As shown in Figure 17, to reach the maximum search throughput, the search latency is approximately 140 ms. On the other hand, to ensure an average latency of < 25 ms, about 1/5 of the maximum search throughput can still be achieved. Such latency is acceptable in RAG workloads because these applications typically require 100ms-level to 1s-level latency for model processing [32, 38]. Similar latency is also observed in the RAG systems [53].

6 Related Work

SSD-resident ANNS. To support ANNS for billion-scale datasets, previous works have used SSDs to avoid storing terabytes of indexes in memory [18, 30, 46, 49, 54]. DiskANN [30] stores the graph index in the SSD, and reduces the search latency by modifying both index-building and searching algorithms. Starling [49] further reduces the number of on-disk I/Os by reshuffling the on-disk index and positioning nearby nodes on the same SSD page. SPANN [18] explores efficient on-disk indexing for cluster-based ANNS by allowing replication of vectors in its nearby clusters. These works focus on latency-critical ANNS scenarios, which fail to provide 100K-scale high throughput due to the limited computing power of CPUs. Instead, we use GPU to exploit multiple SSDs in graph-based ANNS, achieving high I/O efficiency and throughput. FusionANNS [47] leverages GPU for on-disk cluster-based ANNS. However, cluster-based methods are coarse-grained and thus have lower I/O efficiency than graph-based GUSTANN.

ANNS on GPU. Numerous works have explored how to execute graph-based ANNS [26, 41, 48, 57] and cluster-based ANNS [33, 56] on GPU. Recent works have also studied GPU search on larger

datasets. BANG [48] supports graph index for large-scale datasets by coordinating computation on CPU and GPU. RUMMY [56] enables cluster-based FAISS [33] to support large-scale datasets by pipelining the transfers between GPU and DRAM. These works store the index in either DRAM or GPU memory, reducing cost-effectiveness. In addition, GUSTANN optimizes the GPU-based graph search algorithm to address the limited GPU memory and achieve better concurrency.

7 Conclusion

In this paper, we propose a GPU-centric, CPU-assisted architecture for ANNS and present GUST-ANN, a billion-scale ANNS system targeted for high throughput and cost-effectiveness. Experiments show that GUSTANN achieves better throughput and is more cost-effective than current systems.

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References

- [1] 2024. 4th Gen Intel® Xeon® Scalable Processors. <https://ark.intel.com/content/www/us/en/ark/products/series/228622/4th-gen-intel-xeon-scalable-processors.html>.
- [2] 2024. APP Metrics for Intel® Microprocessors. <https://www.intel.com/content/dam/support/us/en/documents/processors/APP-for-Intel-Xeon-Processors.pdf>.
- [3] 2024. Google search statistics 2024 (no. of searches per day). <https://www.demandsage.com/google-search-statistics/>.
- [4] 2024. Introducing OpenAI o1. <https://openai.com/o1/>.
- [5] 2024. New embedding models and API updates. <https://openai.com/index/new-embedding-models-and-api-updates/>.
- [6] 2024. NVIDIA A100 | NVIDIA. <https://www.nvidia.com/en-us/data-center/a100/>.
- [7] 2024. NVIDIA GPUDirect Storage. <https://docs.nvidia.com/gpudirect-storage/index.html>.
- [8] 2024. NVIDIA Tesla A100 Ampere 40 GB Graphics Processor Accelerator - PCIe 4.0 x16 - Dual Slot. <https://www.amazon.com/NVIDIA-Ampere-Graphics-Processor-Accelerator/dp/B08X13X6HF/>.
- [9] 2024. PowerEdge R760 Rack Server. https://www.dell.com/en-us/shop/dell-powered-edge-servers/new-powered-edge-r760-rack-server/spd/poweredge-r760/pe_r760_15724_vi_vp.
- [10] 2024. PowerEdge R760xa Rack Server. <https://www.dell.com/en-us/shop/cty/pdp/spd/poweredge-r760xa/>.
- [11] 2024. Samsung 32GB DDR5 4800MHz PC5-38400 ECC RDIMM 1Rx4 (EC8 10x4) Single Rank 1.1V Registered DIMM 288-Pin Server RAM Memory M321R4GA0BB0-CQK. <https://www.amazon.com/Samsung-4800MHz-PC5-38400-Registered-M321R4GA0BB0-CQK/dp/B0C35566S9>.
- [12] 2024. Samsung PM1743 3.84TB NVMe GEN5 E3.S 1T 12000MBps/12000MBps - MZ3LO3T8HCJR-00A07. <https://www.newegg.com/p/2U3-0005-000N6?Item=9SIA12KKA53814>.
- [13] 2024. Storage Performance Development Kit. <https://spdk.io>.
- [14] 2024. Supermicro 1U SuperStorage Server. https://store.supermicro.com/us_en/1u-superstorage-ssg-121e-ne316r.html.
- [15] Cecilia Aguerrebere, Ishwar Singh Bhati, Mark Hildebrand, Mariano Tepper, and Theodore Willke. 2023. Similarity Search in the Blink of an Eye with Compressed Indices. *Proc. VLDB Endow.* 16, 11 (July 2023), 3433–3446. doi:10.14778/3611479.3611537
- [16] Akhil Arora, Sakshi Sinha, Piyush Kumar, and Arnab Bhattacharya. 2018. HD-index: pushing the scalability-accuracy boundary for approximate kNN search in high-dimensional spaces. *Proc. VLDB Endow.* 11, 8 (apr 2018), 906–919. doi:10.14778/3204028.3204034
- [17] Dmitry Baranchuk, Artem Babenko, and Yury Malkov. 2018. Revisiting the inverted indices for billion-scale approximate nearest neighbors. In *Proceedings of the European Conference on Computer Vision (ECCV)*. 202–216.
- [18] Qi Chen, Bing Zhao, Haidong Wang, Mingqin Li, Chuanjie Liu, Zengzhong Li, Mao Yang, and Jingdong Wang. 2021. SPANN: Highly-efficient Billion-scale Approximate Nearest Neighbor Search. In *35th Conference on Neural Information Processing Systems (NeurIPS 2021)*.
- [19] Rongxin Chen, Yifan Peng, Xingda Wei, Hongrui Xie, Rong Chen, Sijie Shen, and Haibo Chen. 2024. Characterizing the Dilemma of Performance and Index Size in Billion-Scale Vector Search and Breaking It with Second-Tier Memory. *CoRR* abs/2405.03267 (2024). arXiv:2405.03267 doi:10.48550/ARXIV.2405.03267
- [20] Kenneth L. Clarkson. 1994. An Algorithm for Approximate Closest-Point Queries. In *Proceedings of the Tenth Annual Symposium on Computational Geometry, Stony Brook, New York, USA, June 6-8, 1994*, Kurt Mehlhorn (Ed.). ACM, 160–164.